Women Behind Bars:
Gender Inequality and Female Incarceration in Europe

Data Visualisation 2026 - Project Report

Francesco Cocchi, Camilla Cortopassi, Giulio D'Amico, Beatrice Rossi

1. Introduction and Research Question

Crime has historically been studied from a male perspective, with female offending treated as an anomaly. Yet the gendered dimension of criminal justice remains highly relevant. This project investigates the following research question: *how do changes in gender inequality relate to cross-country differences in female incarceration in Europe?*

The question sits at the intersection of sociology, criminology, and data analysis. Two competing theoretical traditions offer contrasting predictions. The emancipation thesis (Adler, 1975) argues that as women gain access to education, employment, and public life, their criminal behaviour will converge with that of men. Conversely, the marginalisation theory (Chesney-Lind, 1986) holds that women enter the criminal justice system not because of empowerment, but because of structural disadvantage: poverty, precarity, and a labour market that was never genuinely equitable.

Earlier frameworks, from Lombroso's biologically deterministic account of the "double deviant" woman (1895) to Heidensohn's 1968 reframing of low female crime rates as a product of social control rather than moral superiority, provide the conceptual foundation for interpreting patterns in contemporary data. This project uses a decade of cross-national European data (2014 - 2023) to test which theoretical perspective the empirical evidence supports.

The project is designed for a non-specialist audience: curious citizens, students, and journalists interested in gender and justice policy. The visual system is intended to be immediately legible, progressively revealing complexity as the user engages more deeply with the data.

2. Data Sources, Integration, and Limitations

The project integrates two publicly available datasets covering all 27 European Union member states over a ten-year period (2014 - 2023), resulting in a panel of 270 country-year observations with no missing values on either key variable.

2.1 Prison Population Data

The primary source for incarceration statistics is Eurostat, which compiles annual data on persons held in penal institutions across EU member states, disaggregated by sex, age class, and nationality. Two core variables were derived: the number of female prisoners per 100,000 inhabitants (PerHundred), computed using annual national population figures from Eurostat, and the female share of the total prison population (% Female), defined as the ratio of female to total prisoners for each country-year observation.

Several countries presented temporal gaps or data inconsistencies in Eurostat. These were resolved using official national sources:

- Germany: Statistisches Bundesamt (Destatis), Strafvollzug series
- Belgium: Rapport annuel, Direction générale des Établissements Pénitentiaires (2014, 2016)
- Ireland: Annual Report 2017, Irish Prison Service
- Portugal: Relatório de Reclusos 2017, Direção-Geral de Reinserção e Serviços Prisionais

All supplementary data were entered manually from official PDF publications after cross-checking against adjacent years for internal consistency.

2.2 Gender Gap Index

Gender equality data were drawn from the World Economic Forum's Global Gender Gap Report, published annually since 2006. The WEF Gender Gap Index scores countries on a 0 to 1 scale across four dimensions: economic participation and opportunity, educational attainment, health and survival, and political empowerment, where 1 represents full parity. Annual index scores for each EU member state were extracted manually from each year's report (2014 - 2023) and compiled into a structured Excel file before merging with the prison dataset.

2.3 Limitations

Several limitations bear on the interpretation of results. First, while all sources are official and institutional, data collection methodologies vary across countries: some report sentenced prisoners only, others include remand populations, and definitions of "prisoner" can differ. This introduces measurement inconsistency that cannot be fully corrected post-hoc.

Second, and more fundamentally, incarceration rates do not reflect the actual volume of crimes committed. This is especially true for women, who are more frequently diverted from custodial sentences through alternative measures such as probation, suspended sentences, or electronic monitoring. The prison population is therefore a downstream indicator shaped by policing practices, prosecutorial discretion, and judicial culture, as well as by underlying offending behaviour.

Third, the analysis is explicitly cross-sectional and exploratory. No causal claims are made. Countries with higher gender equality also differ in welfare state generosity, incarceration philosophy, and legal culture; separating these factors would require multivariate longitudinal modelling beyond the scope of this project.

3. Audience, Tasks, and Design Rationale

The system is designed for a non-expert audience: citizens, students, and policy-interested readers who are curious about the intersection of gender equality and criminal justice, but who lack statistical training. The target user is willing to engage with data if it is presented accessibly and is embedded in a clear narrative.

The visual product is structured around five tasks:

1. **Understand the theoretical debate**: grasp why the relationship between gender equality and female crime is contested.
2. **Explore geographic variation**: compare female incarceration rates across EU member states.
3. **Track change over time**: observe how gender equality has evolved across Europe from 2014 to 2023.
4. **Examine the relationship**: interactively explore the association between the Gender Gap Index and female imprisonment.
5. **Evaluate data quality critically**: understand how the same data can be presented honestly or misleadingly.

Design choices were guided by clarity, coherence, and accessibility. The colour palette draws on EU institutional blue tones, conveying the European geographic scope while maintaining visual consistency. The HTML page follows a linear scrolling narrative: the sociological framing precedes the data, so the user arrives at the visualisations already oriented.

Interactive elements were chosen to support exploration without overwhelming users: the choropleth maps invite geographic comparison immediately, the animated bar chart race shows temporal dynamics in a familiar and engaging format, and the scatterplot allows users to investigate the core

statistical relationship by filtering year and country. All views share a consistent colour scheme, enabling users to transfer what they learn from one chart to the next.

4. System Description

4.1 Overview

The system was built using four tools: Python for data cleaning and exploratory analysis, Tableau, and Power BI for geographic views and animated plots, and HTML/ JavaScript for the final integrated webpage. The webpage constitutes the primary deliverable and embeds all visualisations in a single, self-contained scrollable narrative.

4.2 Visual Components

The system includes eight visual components:

- **Map 1 - Female Prisoners PerHundred**: An interactive map of the EU showing female prisoners per 100,000 inhabitants for each country, built in Python with Plotly, but also with Tableau, and then embedded in the HTML page. Users can hover for exact values and filter by year.
- **Map 2 - Female Prisoners %**: A second interactive map showing the female share of the total prison population. The geographic distribution differs meaningfully from Map 1, revealing how absolute and relative measures can tell different stories.
- **Scatterplot (interactive)**: A Tableau and Power BI scatter plot displaying female prisoners per 100,000 inhabitants against the Gender Gap Index for every country and year. Year and country filters allow users to isolate specific observations.
- **Bar Chart Race (interactive)**: An animated bar chart, made on Tableau and Power BI, showing the evolution of the Gender Gap Index across EU countries from 2014 to 2023. The EU average is highlighted in a contrasting colour using a binary indicator column.
- **Black Hat / White Hat pair (static, Python)**: Two static visualisations presenting the same underlying data in honest and misleading ways respectively. The black hat version uses a truncated y-axis while the white hat version uses a full axis.
- **Exploratory Python charts (supporting analyses)**: Correlation statistics, distribution histograms, and trend lines generated during the EDA phase, some of which were adapted for the final webpage as context-setting figures.

4.3 JavaScript Component

The original JavaScript component consists of a set of interactive visualisations developed directly within the webpage using D3.js. Rather than embedding external outputs, the charts produced during the analysis phase in Python, Tableau, and Power BI were provided as visual references and subsequently reconstructed in D3.js by Claude Design to match the page layout and colour scheme. The result is a set of natively integrated graphics, including the maps and the scatterplot, that respond to user interactions such as hovering, filtering by year, and country selection without any external dependencies or page reloads. Beyond the charts themselves, the JavaScript layer also manages the overall page behaviour: scroll-based transitions between sections and a responsive layout that adapts to different screen sizes. This approach ensures full visual coherence between the narrative structure of the page and the data displays embedded within it.

5. Rejected Alternatives

Several design directions were explored and ultimately set aside.

The first rejected alternative was a prison-corridor concept: the initial idea was to present the data through an immersive metaphor in which the user navigated a virtual prison corridor, with each cell door representing a different EU country. Clicking a door would open a country-level data panel. While conceptually memorable, this approach was abandoned because it privileged the incarceration setting at the expense of the gender equality dimension, which is the analytical core of the project. The navigational metaphor also created usability barriers for non-specialist users unfamiliar with the interactive paradigm.

The second rejected alternative was a rotating globe with zoom-to-Europe animation as the opening visual. This was considered as a dramatic entry point to establish the geographic scope. It was replaced with a static but interactive map for reasons of simplicity, rendering reliability across devices, and reduced load time. The map conveys the same geographic scope without requiring WebGL or heavy JavaScript libraries.

For the webpage itself, WordPress, Google Stich, and Figma were tested before settling on a HTML solution with Claude Design as a drafting assistant. WordPress imposed structural and styling constraints that conflicted with the desired narrative flow; Google Stich offered too little customisation for embedded interactive charts; Figma is a too technical design tool. The HTML approach gave full control over layout, interaction, and visual coherence.

6. Findings, White Hat / Black Hat, and Future Work

6.1 Key Findings

Across 270 country-year observations (27 EU member states, 2014-2023), the data yield two distinct findings depending on how female incarceration is measured.

Female prisoners per 100,000 inhabitants shows a statistically significant negative correlation with the Gender Gap Index (Pearson $r = -0.360$; Spearman $\rho = -0.377$; $p < 0.001$ for both). Countries with higher gender equality consistently imprison fewer women relative to their population. This finding is robust across both linear and rank-based tests, indicating that it is not an artefact of distributional skew or influential outliers.

The female share of total prisoners shows no statistically meaningful association with the Gender Gap Index (Pearson $r = -0.078$, $p = 0.203$; Spearman $\rho = -0.112$, $p = 0.067$). This divergence is substantive, not merely methodological: when gender equality increases, the absolute rate of female imprisonment tends to fall, but this occurs in parallel with comparable shifts in male incarceration, leaving the internal gender ratio of prisons largely unchanged. The driver appears to be a systemic effect on criminality and criminal justice broadly, not a gender-specific one.

These findings directly challenge the emancipation thesis, which predicted that greater equality would raise female crime rates. The evidence points instead toward the marginalisation theory: structural disadvantage, not empowerment, is the primary driver of female contact with criminal justice systems. This aligns with Nordic research (Estrada et al., 2016) showing that the narrowing gender gap in crime is driven by declining male offending, not rising female offending.

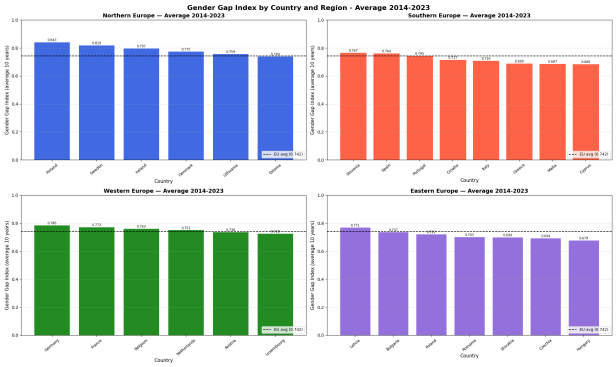
A notable anomaly is Finland, which ranks among the top five countries for female share of total prisoners despite having one of the highest gender equality scores and a very low absolute rate of female incarceration. This reflects the mechanical effect of exceptionally low male incarceration, which inflates the female proportion without any increase in women's absolute imprisonment.

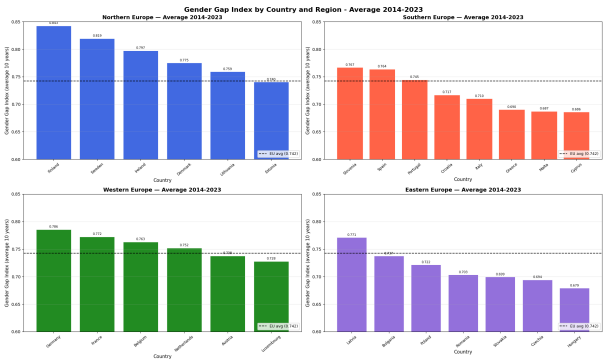
6.2 White Hat / Black Hat Reflection

The black hat and white hat visualisations both display the average Gender Gap Index by country and European region over the period 2014- 2023 but differ in one critical design choice: the y-axis range. The black hat version sets the y-axis floor at 0.60, which compresses the visible range and visually

amplifies the differences between countries. Within each regional panel, gaps that are numerically modest, often just a few hundredths of a point on a 0-to-1 scale, appear dramatic, potentially leading the reader to conclude that gender inequality varies enormously across Europe.

The white hat counterpart uses a full y-axis starting at 0, which correctly contextualises the same values. From this perspective, it becomes immediately clear that all EU member states cluster within a relatively narrow band, and that even the most unequal countries score above 0.67. The contrast between the two versions illustrates a well-known manipulation technique: truncating the axis to exaggerate differences without altering a single data point. It reinforces the importance of axis integrity as a basic principle of honest data communication, particularly when visualisations are used to inform public debate on sensitive topics such as gender policy.





6.3 Future Improvements

The most significant area for improvement lies in data quality. Access to harmonised microdata on charges, sentence lengths, and pre-trial detention would allow a more granular analysis of whether cross-national differences in female incarceration reflect differences in offending, policing, prosecutorial discretion, or judicial practice. The current dataset captures only the downstream outcome.

On the visualisation side, future iterations could implement the rotating globe concept with performance optimisation, add panel regression functionality directly in the interface to allow users to control for confounding country-level variables, and expand the time series to include pre-2014 data where comparable gender equality indices are available.

7. Conclusion

A decade of European data tells a clear story: where women have more, fewer end up behind bars. The association between gender equality and lower female incarceration is statistically robust and geographically consistent, and it runs counter to the intuition that empowerment leads to more crime. If anything, the evidence suggests the opposite: it is disadvantage, not freedom, that pushes women into contact with criminal justice systems.

Numbers alone, however, do not speak for themselves. As the black hat and white hat exercise reminded us, the same data can be made to whisper or to shout depending on the choices made at the design stage. Honest visualisation is not a technical detail; it is an ethical commitment. That is the thread running through this project, from the sociological theories that opened it to the charts that closed it.

Link to our representations

The full project (including the Python notebook, the HTML webpage, raw datasets, and all visual assets) is publicly available on GitHub at: https://github.com/giuliodamico/Gender-Inequality-Index-vs-Female-jail-population.git

Access to our representations on Power BI:

- Race Chart: https://app.powerbi.com/groups/me/reports/5264738c-9f4f-400b-a10d-b859bd605204/bb06b810a72e92a5f0b5?experience=power-bi
- Scatter Plot: https://app.powerbi.com/groups/me/reports/defed693-ec72-44b5-89b6-91b3c3ca434b/49d2c92d45db0687c5ab?experience=power-bi

Access to our representations on Tableau:

- All four Plots: https://public.tableau.com/app/profile/camilla.cortopassi/vizzes

References

Adler, F. (1975). *Sisters in Crime: The Rise of the New Female Criminal*. McGraw-Hill.

Chesney-Lind, M. (1986). Women and Crime: The Female Offender. *Signs: Journal of Women in Culture and Society*, 12(1), 78-96.

Destatis - Statistisches Bundesamt (2023). *Statistischer Bericht: Strafvollzug*. https://www.destatis.de/DE/Themen/Staat/Justiz-Rechtspflege/Publikationen/Downloads-Strafverfolgung-Strafvollzug/statistischer-bericht-strafvollzug-2100410237005.html

Direction générale des Établissements Pénitentiaires - Service public fédéral Justice, Belgique (2014). *Rapport annuel 2014*. https://justice.belgium.be/sites/default/files/downloads/0401_JV_FODJUSTITIE_fr.pdf

Direction générale des Établissements Pénitentiaires - Service public fédéral Justice, Belgique (2016). *Rapport annuel 2016*. https://justice.belgium.be/sites/default/files/bat_ra_2016_fr_light.pdf

Direção-Geral de Reinserção e Serviços Prisionais (2018). *Relatório de Reclusos 2017*. https://dgrsp.justica.gov.pt/Portals/16/Estatisticas/%C3%81rea%20Prisional/Anuais/2017/20180522 11054210Q-reclus-sitpen-sx-nac.pdf

Estrada, F., Björkhagen, A., & Nilsson, A. (2016). The gender gap in offending: Stability and change in Swedish self-report data. *British Journal of Criminology*, 56(6), 1213-1233.

Eurostat (2014-2023). *Prison statistics*. European Commission. https://ec.europa.eu/eurostat/databrowser/view/crim_pris_age$defaultview/default/table?lang=en

Heidensohn, F. (1968). The Deviance of Women: A Critique and an Inquiry. *British Journal of Sociology*, 19(2), 160-175.

Irish Prison Service (2017). *Annual Report 2017*. https://www.irishprisons.ie/wp-content/uploads/documents_pdf/IPS-annualreport-2017.pdf

Lombroso, C. & Ferrero, G. (1895). *La donna delinquente, la prostituta e la donna normale*. Roux.

Pollak, O. (1950). *The Criminality of Women*. University of Pennsylvania Press.

Thomas, W.I. (1923). *The Unadjusted Girl: With Cases and Standpoint for Behavior Analysis*. Little, Brown and Company.

World Economic Forum (2014-2023). *Global Gender Gap Report*. WEF. https://www.weforum.org/reports/global-gender-gap-report-2023

AI Use Appendix

Tool used: Claude (Anthropic), accessed via the Claude.ai interface, specifically the artifact generation feature referred to internally as Claude Design.

The AI tool was used exclusively for the design and development of the final HTML webpage. It was not used for data collection, cleaning, statistical analysis, visualisation creation, or report writing.

The group prepared a detailed prompt structured by page section, specifying the layout, colour scheme, narrative content, and the visualisations to be embedded. Alongside the prompt, the group provided a folder containing all charts produced independently in Python, Tableau, and Power BI, as well as a Word document with the full narrative text. Claude Design used these materials to generate the webpage from scratch, reconstructing the visualisations natively in D3.js to match the page layout and colour scheme.

The prompt was first drafted by the group in Italian, then refined with AI assistance to make it more precise and machine-readable. It specified twelve scrollable sections in order: an opening image with title overlay, an animated timeline illustrating the closing gender gap, the sociological narrative across multiple pages, the methodology section, an interactive European map, a bar chart race, regional area plots, a Power BI view, a heatmap, the black hat visualisation, an interactive quiz, and the white hat visualisation. For each section the group specified the exact content, the source material to use, and any interaction logic required.

The overall structure, layout, and colour palette produced by Claude Design were accepted as the foundation of the final page. However, the group reviewed the full output critically and made direct edits to the HTML file in several areas.